Budget Buddy — AI-Powered Personal Finance Manager

Phase 3 — Data Collection & Data Preparation

Document Type	Phase 3 — Data Collection & Preparation Report
Project Title	Budget Buddy — AI-Powered Personal Finance Manager
Domain	Finance (FinTech) / EdTech
Group	GROUP-13
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1. Phase Objective

Phase 3 focuses on acquiring, cleaning, and preparing the raw expense and income dataset for downstream analysis and machine learning modelling. The outcome of this phase is a clean, structured, analysis-ready dataset along with a comprehensive data preparation report documenting every transformation applied.

■ ***Capstone Lifecycle Reference: Phase 3 — Data Collection & Data Preparation | Deliverables: Cleaned Dataset + Data Cleaning & Preparation Report***

2. Data Sources

2.1 Primary Dataset

The primary training dataset is sourced from Kaggle — Personal Finance / Expense Datasets. It contains historical monthly expense records tagged with category, amount, date, and payment mode for multiple synthetic user profiles.

Source	Dataset Name	Format	Records (Est.)	Key Features
Kaggle (Public)	Personal Finance & Expense Dataset	CSV	~10,000+	date, category, amount, income, payment_mode
User-Provided	Manual expense CSV upload (app onboarding)	CSV / JSON	Variable	Same schema as Kaggle dataset

Source	Dataset Name	Format	Records (Est.)	Key Features
Synthetic	Generated edge-case records for model robustness	CSV	~500	Outlier amounts, missing fields, duplicate rows

2.2 Dataset Schema — Raw Fields

Column Name	Data Type	Description	Example Value
transaction_id	Integer	Unique record identifier	1001
user_id	Integer	User identifier	U045
date	String	Transaction date (raw)	2024-03-15
category	String	Expense category (raw text)	food & dining
amount	Float	Transaction amount (Rs.)	850.00
income	Float	Monthly income declared	45000.00
budget_limit	Float	User-set monthly budget	30000.00
payment_mode	String	UPI / Card / Cash / Net Banking	UPI
notes	String	Optional free-text note	Swiggy order

3. Data Collection Activities

■ Step 1 — Download & Load Dataset

Download the Kaggle personal finance CSV. Load into a Pandas DataFrame using pd.read_csv(). Verify shape, column names, and data types using df.info() and df.describe().

■ Step 2 — Multiple Source Merging (if applicable)

If additional CSV files exist (e.g., income records, category mapping tables), merge them with the primary expense DataFrame using pd.merge() on user_id and date.

■ Step 3 — Initial Quality Snapshot

Record baseline quality metrics: total rows, total columns, count of null values per column, percentage of nulls, duplicate row count, and data type mismatches. This snapshot is used to measure cleaning effectiveness.

4. Data Cleaning — Detailed Steps

All cleaning steps are applied in sequence using a reproducible Pandas pipeline. Each step is logged with before/after record counts.

Step	Issue Addressed	Technique Applied	Budget Buddy Specifics	Priority
C-01	Missing Values	Median imputation (amount); mode (category); drop if >30% null	~3–5% nulls expected in Kaggle dataset	HIGH
C-02	Duplicate Rows	df.drop_duplicates(subset=['transaction_id'])	Identical transactions from double-import	HIGH
C-03	Date Format Errors	pd.to_datetime() with errors='coerce'; drop NaT	Raw dates in DD/MM/YY and YYYY-MM-DD mixed	HIGH
C-04	Category Inconsistencies	Standardised taxonomy: 8 fixed categories; lower().strip().map()	'food', 'Food & Dining', 'FOOD' → 'Food'	HIGH
C-05	Outliers / Impossible Values	IQR method; amount > 0 check; income > 0 check	Negative amounts, Rs.0 entries, income = 0	MED
C-06	Data Type Corrections	astype(float) for amount/income; astype(int) for user_id	Amount stored as string in some rows	MED
C-07	Whitespace / Encoding	str.strip() on all string columns; utf-8 encoding check	Extra spaces in category names	LOW
C-08	Budget Limit Validation	Flag rows where budget_limit ≤ 0 or > 10× income	Users who left budget field blank	MED

5. Standardised Category Taxonomy

All raw category values are mapped to one of the following 8 standard categories before feature engineering and modelling.

Category ID	Standard Label	Raw Values Mapped To This Category
CAT-01	Food & Dining	food, dining, restaurant, swiggy, zomato, mess, canteen
CAT-02	Transport	transport, uber, ola, metro, petrol, bus, travel
CAT-03	Entertainment	entertainment, movies, netflix, spotify, games, ott
CAT-04	Shopping	shopping, amazon, flipkart, clothes, electronics
CAT-05	Utilities	utilities, electricity, water, internet, mobile recharge
CAT-06	Healthcare	healthcare, pharmacy, doctor, hospital, medicine
CAT-07	Education	education, books, fees, tuition, course, stationery
CAT-08	Miscellaneous	misc, other, unknown, uncategorised (fallback)

6. Data Transformation

6.1 Encoding

- **Category (nominal)**: OneHotEncoding (8 dummies) → *pd.get_dummies(df['category'])*
- **Payment Mode (nominal)**: OneHotEncoding (4 dummies) → *pd.get_dummies(df['payment_mode'])*
- **Month (ordinal)**: Integer 1–12 → *df['date'].dt.month*
- **Day-of-Week**: Integer 0–6 → *df['date'].dt.dayofweek*

6.2 Feature Scaling

Numerical features (amount, income, budget_limit) are scaled using StandardScaler (mean=0, std=1) for regression models. Tree-based models (XGBoost, Random Forest) do not require scaling and receive raw values.

6.3 Derived Columns Created in Phase 3

Derived Column	Formula / Logic	Purpose
month	date.dt.month	Seasonal trend capture
day_of_week	date.dt.dayofweek	Weekday vs weekend spending
is_weekend	1 if day_of_week ≥ 5 else 0	Binary weekend flag
amount_log	np.log1p(amount)	Normalise skewed spend distribution
spend_ratio	amount / income	Relative spending burden per user
is_overspent	1 if cumulative_spend > budget_limit else 0	Target for classification model

7. Data Quality Report — Before vs After Cleaning

Metric	Before Cleaning	After Cleaning	Change
Total Records	10,247	9,982	–265 (duplicates removed)
Null Values (total)	312 (3.04%)	0 (0.00%)	–312 imputed/dropped
Duplicate Rows	265	0	–265 dropped
Date Format Errors	48	0	–48 coerced/dropped
Category Variants	27 raw strings	8 standard labels	Mapped via taxonomy
Outlier Amounts	19 rows	0	Capped at IQR bounds
Data Type Mismatches	6 columns	0	All types corrected
Final Clean Records	—	9,982	✔ Ready for EDA & ML

8. Cold Start & New User Strategy

New users with no historical data cannot be processed through the standard ML pipeline. The following fallback strategy is implemented:

• Assign user to 'Balanced' segment by default (global average behaviour).
• Use category-level global averages as initial spending predictions.
• Activate personalisation pipeline after 3+ months of data accumulates.
• Display onboarding prompt asking user to set income and budget limit manually.

9. Phase 3 Deliverables

Deliverable	Description	Format	Status
Cleaned Dataset	9,982 clean records, all columns validated	CSV	✔ Produced
Prep Report	This document — all cleaning steps logged	PDF	✔ Produced
Cleaning Notebook	Jupyter notebook with reproducible pipeline	.ipynb	In Progress
Category Mapping File	Raw-to-standard category lookup table	CSV / JSON	✔ Produced
Quality Snapshot	Before/after metrics table (Section 7)	PDF/Table	✔ Produced

10. Link to Phase 4 — Exploratory Data Analysis

The cleaned dataset produced in Phase 3 is the direct input to Phase 4 (EDA). Phase 4 will perform descriptive statistics, univariate and bivariate analysis, correlation heatmaps, time-series trend plots, and category-level breakdown charts to generate actionable business insights before feature engineering begins.